

HASHAN KAVINGA WEERASOORIYA

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EDUCATION

Purdue University, PhD candidate Intelligent Imaging Lab, Electrical and Computer Engineering	GPA: 3.96/4.00	2022–Present
B.Sc. in Engineering, University of Peradeniya, Sri Lanka. Electrical and Electronic Engineering	GPA: 3.95/4.00	2015–2020

RESEARCH, PUBLICATIONS, AND SELECTED PROJECTS

Single-Photon LiDAR Sensing, Simulation, Compression, and Reconstruction	2023–Present
📖 Single-Photon LiDAR Depth and Reflectivity Estimation for Low-Flux Dynamic Scenes , 2026 (under review). Developed a unified deep-learning framework enabling joint depth and reflectivity reconstruction from photon timestamp frames in fast dynamic scenes. <i>Skills</i> : Deep learning, PyTorch, statistical modeling, inverse problems, simulation	
📖 Single-Photon LiDAR Lab-Bench Construction , <i>Hardware Project</i> , 2024-2025. Designed and implemented a complete single-photon LiDAR laboratory bench integrating TCSPC electronics, single-photon detectors, and galvanometric scanning optics. <i>Skills</i> : Optical system design, data acquisition	
📖 Real-Time Scene-Adaptive Compression for Single-Photon LiDAR , <i>Ongoing Project</i> , 2025–2026. Developing an end-to-end spatiotemporal compression framework deployable on NVIDIA Jetson Xavier, incorporating object tracking and RGB-guided adaptive encoding, while improving 3D reconstruction accuracy using a lightweight neural decoder. <i>Skills</i> : Embedded systems, sensor fusion, compression techniques	
📖 Real-Time Markov Modeling for Single-Photon LiDAR: Accel. and Convergence Analysis , IEEE ICASSP, 2026. Accelerated a Markov-based probabilistic framework for SP-LiDAR, enabling fast generation of dead-time-distorted PDFs. <i>Skills</i> : Probabilistic modeling, Markov processes, algorithm acceleration	
📖 Markov-Renewal Single-Photon LiDAR Simulator , 2026 (under review). Developed a physics-consistent Markov–renewal simulator for single-photon LiDAR that accurately models photon arrivals, detector dead time, and temporal correlations. <i>Skills</i> : Stochastic processes, photon-detection modeling, simulation	
📖 Resolution Limits of Single-Photon LiDAR , <i>IEEE/CVF CVPR</i> , 2024. Analyzed the fundamental trade-off between spatial resolution and per-pixel SNR in array-based single-photon LiDAR systems. <i>Skills</i> : Signal processing, theoretical analysis	
Hyperspectral Imaging for Remote Sensing and Applications	2019–2022
📖 Transmittance Multispectral Imaging for Reheated Coconut Oil Differentiation , <i>IEEE Access</i> , 2022. Designed a cost-effective multispectral imaging system and algorithm to reliably differentiate reheated coconut oils using transmittance spectra. <i>Skills</i> : Optical system design, image processing	
📖 Constrained Nonnegative Matrix Factorization for Blind Hyperspectral Unmixing Incorporating Endmember Independence , <i>IEEE JSTARS</i> , 2021. Developed a hyperspectral matrix factorization framework enforcing endmember independence for improved spectral separation. <i>Skills</i> : Optimization, MATLAB, linear algebra	
📖 Transmittance Multispectral Imaging for Edible Oil Quality Assessment , <i>Optica Imaging and Applied Optics Congress</i> , Vancouver, Canada, 2020. Demonstrated a novel optical imaging prototype for rapid, non-destructive edible oil quality assessment using transmittance spectra. <i>Skills</i> : Optical instrumentation, data acquisition, spectral analysis	
📖 Convolutional Autoencoder for Blind Hyperspectral Image Unmixing , <i>15th IEEE ICIIS</i> , IIT Ropar, India, 2020. Introduced a deep autoencoder for efficient blind hyperspectral unmixing, producing accurate abundance maps and endmembers. <i>Skills</i> : Deep learning, TensorFlow	

EXPERIENCE

Teaching Assistant , Purdue University, USA Assisted graduate and undergraduate courses, including Electrical Engineering Fundamentals I, Digital Signal Processing I, and Machine Learning. Conducted recitations, supervised labs, and provided detailed feedback on assignments, strengthening mentorship and communication skills.	2022–2025
Temporary Instructor , University of Peradeniya, Sri Lanka Instructed and supervised undergraduate communication laboratories; served as a teaching assistant for Network Analysis, Electromagnetic Theory, Digital Signal Processing, Communication Systems, and Telecommunications; and coordinated third-year laboratory scheduling and undergraduate project supervision.	2020–2022

INTERESTS

Signal Processing, Machine Learning, Remote Sensing, Sensor Fusion, Computational Imaging, LiDAR and RADAR

ACHIEVEMENTS

SRC JUMP 2.0 CogniSense Center Best Poster Award for scene-adaptive SP-LiDAR compression.	2025
Prof. E. F. Bartholameusz Endowment Award for the best undergraduate project in Engineering Mathematics.	2020
W. P. Jayasekara Prize for the best undergraduate project in Electrical and Electronic Engineering.	2020